

Expanded Grade 5 Math Practice

Eight Units | 240 Original Questions | Worked Solutions and Answer Key

This workbook follows the eight-unit scope supplied for California Math Expressions Grade 5, Volumes 1 and 2. Questions and explanations in this book are newly written practice material; none are copied from the textbook.

Directions: Show your work. Use drawings, equations, tables, or models where they help. Check answers in the worked-solutions section only after attempting each problem.

Contents

Volume	Chapter	Practice focus	Question
1	1	Multiplication and Division Word Problems	30
1	2	Perimeter and Area	30
1	3	Addition and Subtraction of Whole Numbers and Decimals	30
1	4	Circles, Angles, and Polygons	30
2	5	Addition and Subtraction with Fractions	30
2	6	Volume, Capacity, and Weight	30
2	7	Integers and the Coordinate Plane	30
2	8	Advanced Multiplication and Division with Fractions and Decimals	30

Volume 1

Chapters 1-4: Whole-number operations, measurement, and geometry.

Chapter 1: Multiplication and Division Word Problems

Concept: Multiplicative comparisons and word problems

Use multiplication to find an unknown total or compare quantities. Read for equal groups and phrases such as “times as many.”

Questions

1. A library packs 4 boxes with 11 books in each box. How many books are packed?
2. A library packs 5 boxes with 15 books in each box. How many books are packed?
3. A library packs 6 boxes with 19 books in each box. How many books are packed?
4. A library packs 7 boxes with 23 books in each box. How many books are packed?
5. A library packs 8 boxes with 27 books in each box. How many books are packed?
6. A library packs 9 boxes with 31 books in each box. How many books are packed?
7. A library packs 10 boxes with 35 books in each box. How many books are packed?
8. A library packs 11 boxes with 39 books in each box. How many books are packed?
9. A library packs 12 boxes with 43 books in each box. How many books are packed?
10. A library packs 13 boxes with 47 books in each box. How many books are packed?
11. Maya collected 66 shells. Diego collected 4 times as many. How many shells did Diego collect?
12. Maya collected 72 shells. Diego collected 5 times as many. How many shells did Diego collect?
13. Maya collected 78 shells. Diego collected 6 times as many. How many shells did Diego collect?
14. Maya collected 84 shells. Diego collected 7 times as many. How many shells did Diego collect?

15. Maya collected 90 shells. Diego collected 8 times as many. How many shells did Diego collect?

Concept: Expressions, parentheses, and properties

Parentheses tell you what to calculate first. The distributive property breaks one factor into easier parts.

Questions

16. Evaluate $(20 + 48) \times 7$.

17. Evaluate $(21 + 51) \times 8$.

18. Evaluate $(22 + 54) \times 9$.

19. Evaluate $(23 + 57) \times 10$.

20. Evaluate $(24 + 60) \times 11$.

21. Use the distributive property to find 11×23 by breaking 23 into $21 + 2$.

22. Use the distributive property to find 12×24 by breaking 24 into $22 + 2$.

23. Use the distributive property to find 13×25 by breaking 25 into $23 + 2$.

24. Use the distributive property to find 14×26 by breaking 26 into $24 + 2$.

25. Use the distributive property to find 15×27 by breaking 27 into $25 + 2$.

Concept: Factors

Factors are whole numbers that multiply to make a product. Find them by making multiplication pairs.

Questions

26. List every factor of 78.

27. List every factor of 81.

28. List every factor of 84.

29. List every factor of 87.

30. List every factor of 90.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

1.1 Answer: 44

Multiply the equal groups: $4 \times 11 = 44$.

1.2 Answer: 75

Multiply the equal groups: $5 \times 15 = 75$.

1.3 Answer: 114

Multiply the equal groups: $6 \times 19 = 114$.

1.4 Answer: 161

Multiply the equal groups: $7 \times 23 = 161$.

1.5 Answer: 216

Multiply the equal groups: $8 \times 27 = 216$.

1.6 Answer: 279

Multiply the equal groups: $9 \times 31 = 279$.

1.7 Answer: 350

Multiply the equal groups: $10 \times 35 = 350$.

1.8 Answer: 429

Multiply the equal groups: $11 \times 39 = 429$.

1.9 Answer: 516

Multiply the equal groups: $12 \times 43 = 516$.

1.10 Answer: 611

Multiply the equal groups: $13 \times 47 = 611$.

1.11 Answer: 264

A multiplicative comparison uses multiplication: $66 \times 4 = 264$.

1.12 Answer: 360

A multiplicative comparison uses multiplication: $72 \times 5 = 360$.

1.13 Answer: 468

A multiplicative comparison uses multiplication: $78 \times 6 = 468$.

1.14 Answer: 588

A multiplicative comparison uses multiplication: $84 \times 7 = 588$.

1.15 Answer: 720

A multiplicative comparison uses multiplication: $90 \times 8 = 720$.

1.16 Answer: 476

Work inside parentheses first: $20 + 48 = 68$. Then multiply: $68 \times 7 = 476$.

1.17 Answer: 576

Work inside parentheses first: $21 + 51 = 72$. Then multiply: $72 \times 8 = 576$.

1.18 Answer: 684

Work inside parentheses first: $22 + 54 = 76$. Then multiply: $76 \times 9 = 684$.

1.19 Answer: 800

Work inside parentheses first: $23 + 57 = 80$. Then multiply: $80 \times 10 = 800$.

1.20 Answer: 924

Work inside parentheses first: $24 + 60 = 84$. Then multiply: $84 \times 11 = 924$.

1.21 Answer: 253

$11 \times (21 + 2) = (11 \times 21) + (11 \times 2) = 231 + 22 = 253$.

1.22 Answer: 288

$12 \times (22 + 2) = (12 \times 22) + (12 \times 2) = 264 + 24 = 288$.

1.23 Answer: 325

$13 \times (23 + 2) = (13 \times 23) + (13 \times 2) = 299 + 26 = 325$.

1.24 Answer: 364

$14 \times (24 + 2) = (14 \times 24) + (14 \times 2) = 336 + 28 = 364$.

1.25 Answer: 405

$15 \times (25 + 2) = (15 \times 25) + (15 \times 2) = 375 + 30 = 405$.

1.26 Answer: 1, 2, 3, 6, 13, 26, 39, 78

Pair factors that multiply to 78: 1×78 , 2×39 , 3×26 , 6×13 .

1.27 Answer: 1, 3, 9, 27, 81

Pair factors that multiply to 81: 1×81 , 3×27 , 9×9 .

1.28 Answer: 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84

Pair factors that multiply to 84: 1×84 , 2×42 , 3×28 , 4×21 , 6×14 , 7×12 .

1.29 Answer: 1, 3, 29, 87

Pair factors that multiply to 87: 1×87 , 3×29 .

1.30 Answer: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90

Pair factors that multiply to 90: 1×90 , 2×45 , 3×30 , 5×18 , 6×15 , 9×10 .

Chapter 2: Perimeter and Area

Concept: Perimeter and area of rectangles

Perimeter measures distance around a shape. Area measures the square units that cover its inside.

Questions

1. A rectangle is 6 cm long and 3 cm wide. Find its perimeter and area.
2. A rectangle is 7 cm long and 4 cm wide. Find its perimeter and area.
3. A rectangle is 8 cm long and 5 cm wide. Find its perimeter and area.
4. A rectangle is 9 cm long and 6 cm wide. Find its perimeter and area.
5. A rectangle is 10 cm long and 7 cm wide. Find its perimeter and area.
6. A rectangle is 11 cm long and 8 cm wide. Find its perimeter and area.
7. A rectangle is 12 cm long and 9 cm wide. Find its perimeter and area.
8. A rectangle is 13 cm long and 10 cm wide. Find its perimeter and area.
9. A rectangle is 14 cm long and 11 cm wide. Find its perimeter and area.
10. A rectangle is 15 cm long and 12 cm wide. Find its perimeter and area.

Concept: Finding an unknown side length

When area and one side are known, divide the area by the known side to find the missing side.

Questions

11. A garden has an area of 126 square meters and a width of 7 meters. What is its length?
12. A garden has an area of 152 square meters and a width of 8 meters. What is its length?
13. A garden has an area of 180 square meters and a width of 9 meters. What is its length?

14. A garden has an area of 210 square meters and a width of 10 meters. What is its length?
15. A garden has an area of 242 square meters and a width of 11 meters. What is its length?
16. A garden has an area of 276 square meters and a width of 12 meters. What is its length?
17. A garden has an area of 312 square meters and a width of 13 meters. What is its length?
18. A garden has an area of 350 square meters and a width of 14 meters. What is its length?
19. A garden has an area of 390 square meters and a width of 15 meters. What is its length?
20. A garden has an area of 432 square meters and a width of 16 meters. What is its length?

Concept: Area models for multiplication

Split a two-digit factor into tens and ones, multiply each part, and add the partial products.

Questions

21. Use an area model or the distributive property to find 111×7 .
22. Use an area model or the distributive property to find 124×8 .
23. Use an area model or the distributive property to find 137×9 .
24. Use an area model or the distributive property to find 141×10 .
25. Use an area model or the distributive property to find 154×11 .
26. Use an area model or the distributive property to find 167×12 .
27. Use an area model or the distributive property to find 171×13 .
28. Use an area model or the distributive property to find 184×14 .
29. Use an area model or the distributive property to find 197×15 .

30. Use an area model or the distributive property to find 201×16 .

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

2.1 Answer: Perimeter = 18 cm; area = 18 sq cm

Perimeter = $2 \times (6 + 3) = 18$ cm. Area = $6 \times 3 = 18$ square centimeters.

2.2 Answer: Perimeter = 22 cm; area = 28 sq cm

Perimeter = $2 \times (7 + 4) = 22$ cm. Area = $7 \times 4 = 28$ square centimeters.

2.3 Answer: Perimeter = 26 cm; area = 40 sq cm

Perimeter = $2 \times (8 + 5) = 26$ cm. Area = $8 \times 5 = 40$ square centimeters.

2.4 Answer: Perimeter = 30 cm; area = 54 sq cm

Perimeter = $2 \times (9 + 6) = 30$ cm. Area = $9 \times 6 = 54$ square centimeters.

2.5 Answer: Perimeter = 34 cm; area = 70 sq cm

Perimeter = $2 \times (10 + 7) = 34$ cm. Area = $10 \times 7 = 70$ square centimeters.

2.6 Answer: Perimeter = 38 cm; area = 88 sq cm

Perimeter = $2 \times (11 + 8) = 38$ cm. Area = $11 \times 8 = 88$ square centimeters.

2.7 Answer: Perimeter = 42 cm; area = 108 sq cm

Perimeter = $2 \times (12 + 9) = 42$ cm. Area = $12 \times 9 = 108$ square centimeters.

2.8 Answer: Perimeter = 46 cm; area = 130 sq cm

Perimeter = $2 \times (13 + 10) = 46$ cm. Area = $13 \times 10 = 130$ square centimeters.

2.9 Answer: Perimeter = 50 cm; area = 154 sq cm

Perimeter = $2 \times (14 + 11) = 50$ cm. Area = $14 \times 11 = 154$ square centimeters.

2.10 Answer: Perimeter = 54 cm; area = 180 sq cm

Perimeter = $2 \times (15 + 12) = 54$ cm. Area = $15 \times 12 = 180$ square centimeters.

2.11 Answer: 18 m

Area = length $\times$ width, so length = 126 divided by 7 = 18 meters.

2.12 Answer: 19 m

Area = length $\times$ width, so length = 152 divided by 8 = 19 meters.

2.13 Answer: 20 m

Area = length $\times$ width, so length = 180 divided by 9 = 20 meters.

2.14 Answer: 21 m

Area = length $\times$ width, so length = 210 divided by 10 = 21 meters.

2.15 Answer: 22 m

Area = length $\times$ width, so length = 242 divided by 11 = 22 meters.

2.16 Answer: 23 m

Area = length $\times$ width, so length = 276 divided by 12 = 23 meters.

2.17 Answer: 24 m

Area = length $\times$ width, so length = 312 divided by 13 = 24 meters.

2.18 Answer: 25 m

Area = length $\times$ width, so length = 350 divided by 14 = 25 meters.

2.19 Answer: 26 m

Area = length x width, so length = 390 divided by 15 = 26 meters.

2.20 Answer: 27 m

Area = length x width, so length = 432 divided by 16 = 27 meters.

2.21 Answer: 777

Break 111 into 110 + 1: $(110 \times 7) + (1 \times 7) = 770 + 7 = 777$.

2.22 Answer: 992

Break 124 into 120 + 4: $(120 \times 8) + (4 \times 8) = 960 + 32 = 992$.

2.23 Answer: 1233

Break 137 into 130 + 7: $(130 \times 9) + (7 \times 9) = 1170 + 63 = 1233$.

2.24 Answer: 1410

Break 141 into 140 + 1: $(140 \times 10) + (1 \times 10) = 1400 + 10 = 1410$.

2.25 Answer: 1694

Break 154 into 150 + 4: $(150 \times 11) + (4 \times 11) = 1650 + 44 = 1694$.

2.26 Answer: 2004

Break 167 into 160 + 7: $(160 \times 12) + (7 \times 12) = 1920 + 84 = 2004$.

2.27 Answer: 2223

Break 171 into 170 + 1: $(170 \times 13) + (1 \times 13) = 2210 + 13 = 2223$.

2.28 Answer: 2576

Break 184 into 180 + 4: $(180 \times 14) + (4 \times 14) = 2520 + 56 = 2576$.

2.29 Answer: 2955

Break 197 into 190 + 7: $(190 \times 15) + (7 \times 15) = 2850 + 105 = 2955$.

2.30 Answer: 3216

Break 201 into 200 + 1: $(200 \times 16) + (1 \times 16) = 3200 + 16 = 3216$.

Chapter 3: Addition and Subtraction of Whole Numbers and Decimals

Concept: Adding and subtracting whole numbers

Line up digits by place value. Regroup when a column has too many ones or too few to subtract.

Questions

1. Find $27,321 + 13,109$.
2. Find $31,642 + 15,218$.
3. Find $35,963 + 17,327$.
4. Find $40,284 + 19,436$.
5. Find $44,605 + 21,545$.
6. Find $48,926 + 23,654$.
7. Find $53,247 + 25,763$.
8. Find $57,568 + 27,872$.
9. Find $101,853 - 38,569$.
10. Find $103,170 - 39,410$.
11. Find $104,487 - 40,251$.
12. Find $105,804 - 41,092$.
13. Find $107,121 - 41,933$.
14. Find $108,438 - 42,774$.
15. Find $109,755 - 43,615$.

Concept: Adding and subtracting decimals

Always line up decimal points. Add zeros as placeholders when a number has fewer decimal places.

Questions

16. Find $19.47 + 2.18$.

17. Find $20.47 + 2.28$.

18. Find $21.47 + 2.38$.

19. Find $22.47 + 2.48$.

20. Find $23.47 + 2.58$.

21. Find $24.47 + 2.68$.

22. Find $25.47 + 2.78$.

23. Find $26.47 + 2.88$.

24. Find $32.36 - 3.87$.

25. Find $33.36 - 3.97$.

26. Find $34.36 - 4.07$.

27. Find $35.36 - 4.17$.

28. Find $36.36 - 4.27$.

29. Find $37.36 - 4.37$.

30. Find $38.36 - 4.47$.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

3.1 Answer: 40430

Align place values and add: $27,321 + 13,109 = 40,430$.

3.2 Answer: 46860

Align place values and add: $31,642 + 15,218 = 46,860$.

3.3 Answer: 53290

Align place values and add: $35,963 + 17,327 = 53,290$.

3.4 Answer: 59720

Align place values and add: $40,284 + 19,436 = 59,720$.

3.5 Answer: 66150

Align place values and add: $44,605 + 21,545 = 66,150$.

3.6 Answer: 72580

Align place values and add: $48,926 + 23,654 = 72,580$.

3.7 Answer: 79010

Align place values and add: $53,247 + 25,763 = 79,010$.

3.8 Answer: 85440

Align place values and add: $57,568 + 27,872 = 85,440$.

3.9 Answer: 63284

Align place values and subtract: $101,853 - 38,569 = 63,284$.

3.10 Answer: 63760

Align place values and subtract: $103,170 - 39,410 = 63,760$.

3.11 Answer: 64236

Align place values and subtract: $104,487 - 40,251 = 64,236$.

3.12 Answer: 64712

Align place values and subtract: $105,804 - 41,092 = 64,712$.

3.13 Answer: 65188

Align place values and subtract: $107,121 - 41,933 = 65,188$.

3.14 Answer: 65664

Align place values and subtract: $108,438 - 42,774 = 65,664$.

3.15 Answer: 66140

Align place values and subtract: $109,755 - 43,615 = 66,140$.

3.16 Answer: 21.65

Line up the decimal points: $19.47 + 2.18 = 21.65$.

3.17 Answer: 22.75

Line up the decimal points: $20.47 + 2.28 = 22.75$.

3.18 Answer: 23.85

Line up the decimal points: $21.47 + 2.38 = 23.85$.

3.19 Answer: 24.95

Line up the decimal points: $22.47 + 2.48 = 24.95$.

3.20 Answer: 26.05

Line up the decimal points: $23.47 + 2.58 = 26.05$.

3.21 Answer: 27.15

Line up the decimal points: $24.47 + 2.68 = 27.15$.

3.22 Answer: 28.25

Line up the decimal points: $25.47 + 2.78 = 28.25$.

3.23 Answer: 29.35

Line up the decimal points: $26.47 + 2.88 = 29.35$.

3.24 Answer: 28.49

Line up the decimal points: $32.36 - 3.87 = 28.49$.

3.25 Answer: 29.39

Line up the decimal points: $33.36 - 3.97 = 29.39$.

3.26 Answer: 30.29

Line up the decimal points: $34.36 - 4.07 = 30.29$.

3.27 Answer: 31.19

Line up the decimal points: $35.36 - 4.17 = 31.19$.

3.28 Answer: 32.09

Line up the decimal points: $36.36 - 4.27 = 32.09$.

3.29 Answer: 32.99

Line up the decimal points: $37.36 - 4.37 = 32.99$.

3.30 Answer: 33.89

Line up the decimal points: $38.36 - 4.47 = 33.89$.

Chapter 4: Circles, Angles, and Polygons

Concept: Measuring and classifying angles

Acute angles are less than 90 degrees, right angles equal 90 degrees, obtuse angles are between 90 and 180 degrees, and straight angles equal 180 degrees.

Questions

1. Classify an angle that measures 25 degrees.
2. Classify an angle that measures 48 degrees.
3. Classify an angle that measures 72 degrees.
4. Classify an angle that measures 90 degrees.
5. Classify an angle that measures 105 degrees.
6. Classify an angle that measures 128 degrees.
7. Classify an angle that measures 160 degrees.
8. Classify an angle that measures 180 degrees.
9. Classify an angle that measures 37 degrees.
10. Classify an angle that measures 145 degrees.

Concept: Angles on a straight line

The two adjacent angles on a straight line total 180 degrees. Subtract a known angle from 180 to find the other.

Questions

11. Two angles form a straight line. One measures 97 degrees. What is the other angle?
12. Two angles form a straight line. One measures 104 degrees. What is the other angle?
13. Two angles form a straight line. One measures 111 degrees. What is the other angle?

14. Two angles form a straight line. One measures 118 degrees. What is the other angle?
15. Two angles form a straight line. One measures 125 degrees. What is the other angle?
16. Two angles form a straight line. One measures 132 degrees. What is the other angle?
17. Two angles form a straight line. One measures 139 degrees. What is the other angle?
18. Two angles form a straight line. One measures 146 degrees. What is the other angle?
19. Two angles form a straight line. One measures 153 degrees. What is the other angle?
20. Two angles form a straight line. One measures 160 degrees. What is the other angle?

Concept: Polygons

A polygon is a closed shape made from line segments. Its number of sides equals its number of vertices.

Questions

21. How many sides and vertices does a triangle have?
22. How many sides and vertices does a quadrilateral have?
23. How many sides and vertices does a pentagon have?
24. How many sides and vertices does a hexagon have?
25. How many sides and vertices does a heptagon have?
26. How many sides and vertices does a octagon have?
27. How many sides and vertices does a nonagon have?
28. How many sides and vertices does a decagon have?
29. How many sides and vertices does a square have?

30. How many sides and vertices does an equilateral triangle have?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

4.1 Answer: acute

25 degrees is acute because it is less than 90 degrees.

4.2 Answer: acute

48 degrees is acute because it is less than 90 degrees.

4.3 Answer: acute

72 degrees is acute because it is less than 90 degrees.

4.4 Answer: right

90 degrees is right because it is equal to 90 degrees.

4.5 Answer: obtuse

105 degrees is obtuse because it is between 90 and 180 degrees.

4.6 Answer: obtuse

128 degrees is obtuse because it is between 90 and 180 degrees.

4.7 Answer: obtuse

160 degrees is obtuse because it is between 90 and 180 degrees.

4.8 Answer: straight

180 degrees is straight because it is equal to 180 degrees.

4.9 Answer: acute

37 degrees is acute because it is less than 90 degrees.

4.10 Answer: obtuse

145 degrees is obtuse because it is between 90 and 180 degrees.

4.11 Answer: 83 degrees

Angles on a straight line total 180 degrees: $180 - 97 = 83$ degrees.

4.12 Answer: 76 degrees

Angles on a straight line total 180 degrees: $180 - 104 = 76$ degrees.

4.13 Answer: 69 degrees

Angles on a straight line total 180 degrees: $180 - 111 = 69$ degrees.

4.14 Answer: 62 degrees

Angles on a straight line total 180 degrees: $180 - 118 = 62$ degrees.

4.15 Answer: 55 degrees

Angles on a straight line total 180 degrees: $180 - 125 = 55$ degrees.

4.16 Answer: 48 degrees

Angles on a straight line total 180 degrees: $180 - 132 = 48$ degrees.

4.17 Answer: 41 degrees

Angles on a straight line total 180 degrees: $180 - 139 = 41$ degrees.

4.18 Answer: 34 degrees

Angles on a straight line total 180 degrees: $180 - 146 = 34$ degrees.

4.19 Answer: 27 degrees

Angles on a straight line total 180 degrees: $180 - 153 = 27$ degrees.

4.20 Answer: 20 degrees

Angles on a straight line total 180 degrees: $180 - 160 = 20$ degrees.

4.21 Answer: 3 sides and 3 vertices

A triangle has 3 sides. Each corner is a vertex, so it has 3 vertices.

4.22 Answer: 4 sides and 4 vertices

A quadrilateral has 4 sides. Each corner is a vertex, so it has 4 vertices.

4.23 Answer: 5 sides and 5 vertices

A pentagon has 5 sides. Each corner is a vertex, so it has 5 vertices.

4.24 Answer: 6 sides and 6 vertices

A hexagon has 6 sides. Each corner is a vertex, so it has 6 vertices.

4.25 Answer: 7 sides and 7 vertices

A heptagon has 7 sides. Each corner is a vertex, so it has 7 vertices.

4.26 Answer: 8 sides and 8 vertices

An octagon has 8 sides. Each corner is a vertex, so it has 8 vertices.

4.27 Answer: 9 sides and 9 vertices

A nonagon has 9 sides. Each corner is a vertex, so it has 9 vertices.

4.28 Answer: 10 sides and 10 vertices

A decagon has 10 sides. Each corner is a vertex, so it has 10 vertices.

4.29 Answer: 4 sides and 4 vertices

A square has 4 sides. Each corner is a vertex, so it has 4 vertices.

4.30 Answer: 3 sides and 3 vertices

An equilateral triangle has 3 sides. Each corner is a vertex, so it has 3 vertices.

Volume 2

Chapters 5-8: Fractions, volume, coordinates, and advanced operations.

Chapter 5: Addition and Subtraction with Fractions

Concept: Equivalent fractions

Equivalent fractions name the same amount. Multiply the numerator and denominator by the same nonzero number.

Questions

1. Write an equivalent fraction for $\frac{1}{2}$ with denominator 4.
2. Write an equivalent fraction for $\frac{2}{3}$ with denominator 6.
3. Write an equivalent fraction for $\frac{3}{4}$ with denominator 8.
4. Write an equivalent fraction for $\frac{3}{5}$ with denominator 10.
5. Write an equivalent fraction for $\frac{4}{7}$ with denominator 14.
6. Write an equivalent fraction for $\frac{5}{8}$ with denominator 16.
7. Write an equivalent fraction for $\frac{2}{9}$ with denominator 18.
8. Write an equivalent fraction for $\frac{3}{10}$ with denominator 20.

Concept: Adding fractions

Use a common denominator so the pieces are the same size, then add the numerators.

Questions

9. Find $\frac{1}{3} + \frac{1}{6}$.
10. Find $\frac{2}{5} + \frac{1}{10}$.
11. Find $\frac{3}{4} + \frac{1}{8}$.
12. Find $\frac{2}{3} + \frac{1}{9}$.

13. Find $\frac{3}{5} + \frac{1}{4}$.

14. Find $\frac{5}{6} + \frac{1}{3}$.

15. Find $\frac{7}{8} + \frac{1}{4}$.

Concept: Subtracting fractions

Rewrite fractions with a common denominator, subtract the numerators, and simplify the result when possible.

Questions

16. Find $\frac{4}{7} - \frac{2}{7}$.

17. Find $\frac{5}{9} - \frac{2}{3}$.

18. Find $\frac{7}{10} - \frac{3}{5}$.

19. Find $\frac{11}{12} - \frac{1}{6}$.

20. Find $\frac{3}{4} - \frac{2}{5}$.

21. Find $\frac{5}{6} - \frac{1}{2}$.

22. Find $\frac{7}{9} - \frac{1}{3}$.

Concept: Comparing fractions and decimals

Convert to a shared form, such as decimals, or compare each number to useful benchmarks like one-half and one.

Questions

23. Compare $\frac{3}{4}$ and 0.58. Use $<$, $>$, or $=$.

24. Compare $\frac{5}{8}$ and 0.70. Use $<$, $>$, or $=$.

25. Compare $\frac{7}{12}$ and 0.60. Use $<$, $>$, or $=$.

26. Compare $\frac{9}{10}$ and 0.90. Use $<$, $>$, or $=$.

27. Compare $\frac{4}{9}$ and 0.42. Use $<$, $>$, or $=$.

28. Compare $\frac{3}{5}$ and 0.55. Use $<$, $>$, or $=$.

29. Compare $\frac{7}{8}$ and 0.80. Use $<$, $>$, or $=$.

30. Compare $\frac{2}{5}$ and 0.45. Use $<$, $>$, or $=$.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

5.1 Answer: $\frac{2}{4}$

Multiply numerator and denominator by 2: $\frac{1}{2} = \frac{2}{4}$.

5.2 Answer: $\frac{4}{6}$

Multiply numerator and denominator by 2: $\frac{2}{3} = \frac{4}{6}$.

5.3 Answer: $\frac{6}{8}$

Multiply numerator and denominator by 2: $\frac{3}{4} = \frac{6}{8}$.

5.4 Answer: $\frac{6}{10}$

Multiply numerator and denominator by 2: $\frac{3}{5} = \frac{6}{10}$.

5.5 Answer: $\frac{8}{14}$

Multiply numerator and denominator by 2: $\frac{4}{7} = \frac{8}{14}$.

5.6 Answer: $\frac{10}{16}$

Multiply numerator and denominator by 2: $\frac{5}{8} = \frac{10}{16}$.

5.7 Answer: $\frac{4}{18}$

Multiply numerator and denominator by 2: $\frac{2}{9} = \frac{4}{18}$.

5.8 Answer: $\frac{6}{20}$

Multiply numerator and denominator by 2: $\frac{3}{10} = \frac{6}{20}$.

5.9 Answer: $\frac{1}{2}$

Use a common denominator, then add. The sum simplifies to $\frac{1}{2}$.

5.10 Answer: $\frac{1}{2}$

Use a common denominator, then add. The sum simplifies to $\frac{1}{2}$.

5.11 Answer: $\frac{7}{8}$

Use a common denominator, then add. The sum simplifies to $\frac{7}{8}$.

5.12 Answer: $\frac{7}{9}$

Use a common denominator, then add. The sum simplifies to $\frac{7}{9}$.

5.13 Answer: $\frac{17}{20}$

Use a common denominator, then add. The sum simplifies to $\frac{17}{20}$.

5.14 Answer: $1\frac{1}{6}$

Use a common denominator, then add. The sum simplifies to $1\frac{1}{6}$.

5.15 Answer: $1 \frac{1}{8}$

Use a common denominator, then add. The sum simplifies to $1 \frac{1}{8}$.

5.16 Answer: $\frac{2}{7}$

Use a common denominator, then subtract. The difference simplifies to $\frac{2}{7}$.

5.17 Answer: $-1 \frac{8}{9}$

Use a common denominator, then subtract. The difference simplifies to $-1 \frac{8}{9}$.

5.18 Answer: $\frac{1}{10}$

Use a common denominator, then subtract. The difference simplifies to $\frac{1}{10}$.

5.19 Answer: $\frac{3}{4}$

Use a common denominator, then subtract. The difference simplifies to $\frac{3}{4}$.

5.20 Answer: $\frac{7}{20}$

Use a common denominator, then subtract. The difference simplifies to $\frac{7}{20}$.

5.21 Answer: $\frac{1}{3}$

Use a common denominator, then subtract. The difference simplifies to $\frac{1}{3}$.

5.22 Answer: $\frac{4}{9}$

Use a common denominator, then subtract. The difference simplifies to $\frac{4}{9}$.

5.23 Answer: $>$

Convert $\frac{3}{4}$ to a decimal or use a common benchmark. $0.75 > 0.58$.

5.24 Answer: $<$

Convert $\frac{5}{8}$ to a decimal or use a common benchmark. $0.625 < 0.70$.

5.25 Answer: $<$

Convert $\frac{7}{12}$ to a decimal or use a common benchmark. $0.583 < 0.60$.

5.26 Answer: $<$

Convert $\frac{9}{10}$ to a decimal or use a common benchmark. $0.9 < 0.90$.

5.27 Answer: $>$

Convert $\frac{4}{9}$ to a decimal or use a common benchmark. $0.444 > 0.42$.

5.28 Answer: $>$

Convert $\frac{3}{5}$ to a decimal or use a common benchmark. $0.6 > 0.55$.

5.29 Answer: $>$

Convert $\frac{7}{8}$ to a decimal or use a common benchmark. $0.875 > 0.80$.

5.30 Answer: $<$

Convert $\frac{2}{5}$ to a decimal or use a common benchmark. $0.4 < 0.45$.

Chapter 6: Volume, Capacity, and Weight

Concept: Volume of rectangular prisms

Volume is the number of cubic units that fill a solid. Multiply length, width, and height: $V = l \times w \times h$.

Questions

1. Find the volume of a rectangular prism with length 3 cm, width 4 cm, and height 2 cm.
2. Find the volume of a rectangular prism with length 4 cm, width 5 cm, and height 3 cm.
3. Find the volume of a rectangular prism with length 5 cm, width 6 cm, and height 4 cm.
4. Find the volume of a rectangular prism with length 6 cm, width 7 cm, and height 5 cm.
5. Find the volume of a rectangular prism with length 7 cm, width 8 cm, and height 6 cm.
6. Find the volume of a rectangular prism with length 8 cm, width 9 cm, and height 7 cm.
7. Find the volume of a rectangular prism with length 9 cm, width 10 cm, and height 8 cm.
8. Find the volume of a rectangular prism with length 10 cm, width 11 cm, and height 9 cm.
9. Find the volume of a rectangular prism with length 11 cm, width 12 cm, and height 10 cm.
10. Find the volume of a rectangular prism with length 12 cm, width 13 cm, and height 11 cm.

Concept: Capacity, mass, and length conversions

Use a conversion relationship to change units. Multiply when moving from a larger unit to a smaller unit.

Questions

11. Convert 3 feet to inches.
12. Convert 5 yards to feet.
13. Convert 4 pints to cups.

14. Convert 3 quarts to pints.
15. Convert 2 gallons to quarts.
16. Convert 6 pounds to ounces.
17. Convert 7 kilograms to grams.
18. Convert 9 meters to centimeters.
19. Convert 4 liters to milliliters.
20. Convert 8 tons to pounds.

Concept: Volume as equal layers

A rectangular prism has equal layers. Multiply the cubes in one layer by the number of layers.

Questions

21. A box has 7 equal layers of unit cubes. Each layer has 25 cubes. What is the volume?
22. A box has 8 equal layers of unit cubes. Each layer has 26 cubes. What is the volume?
23. A box has 9 equal layers of unit cubes. Each layer has 27 cubes. What is the volume?
24. A box has 10 equal layers of unit cubes. Each layer has 28 cubes. What is the volume?
25. A box has 11 equal layers of unit cubes. Each layer has 29 cubes. What is the volume?
26. A box has 12 equal layers of unit cubes. Each layer has 30 cubes. What is the volume?
27. A box has 13 equal layers of unit cubes. Each layer has 31 cubes. What is the volume?
28. A box has 14 equal layers of unit cubes. Each layer has 32 cubes. What is the volume?
29. A box has 15 equal layers of unit cubes. Each layer has 33 cubes. What is the volume?

30. A box has 16 equal layers of unit cubes. Each layer has 34 cubes. What is the volume?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

6.1 Answer: 24 cubic cm

Volume = $l \times w \times h = 3 \times 4 \times 2 = 24$ cubic centimeters.

6.2 Answer: 60 cubic cm

Volume = $l \times w \times h = 4 \times 5 \times 3 = 60$ cubic centimeters.

6.3 Answer: 120 cubic cm

Volume = $l \times w \times h = 5 \times 6 \times 4 = 120$ cubic centimeters.

6.4 Answer: 210 cubic cm

Volume = $l \times w \times h = 6 \times 7 \times 5 = 210$ cubic centimeters.

6.5 Answer: 336 cubic cm

Volume = $l \times w \times h = 7 \times 8 \times 6 = 336$ cubic centimeters.

6.6 Answer: 504 cubic cm

Volume = $l \times w \times h = 8 \times 9 \times 7 = 504$ cubic centimeters.

6.7 Answer: 720 cubic cm

Volume = $l \times w \times h = 9 \times 10 \times 8 = 720$ cubic centimeters.

6.8 Answer: 990 cubic cm

Volume = $l \times w \times h = 10 \times 11 \times 9 = 990$ cubic centimeters.

6.9 Answer: 1320 cubic cm

Volume = $l \times w \times h = 11 \times 12 \times 10 = 1320$ cubic centimeters.

6.10 Answer: 1716 cubic cm

Volume = $l \times w \times h = 12 \times 13 \times 11 = 1716$ cubic centimeters.

6.11 Answer: 36 inches

Use the conversion relationship: 3 feet = 36 inches.

6.12 Answer: 15 feet

Use the conversion relationship: 5 yards = 15 feet.

6.13 Answer: 8 cups

Use the conversion relationship: 4 pints = 8 cups.

6.14 Answer: 6 pints

Use the conversion relationship: 3 quarts = 6 pints.

6.15 Answer: 8 quarts

Use the conversion relationship: 2 gallons = 8 quarts.

6.16 Answer: 96 ounces

Use the conversion relationship: 6 pounds = 96 ounces.

6.17 Answer: 7000 grams

Use the conversion relationship: 7 kilograms = 7000 grams.

6.18 Answer: 900 centimeters

Use the conversion relationship: 9 meters = 900 centimeters.

6.19 Answer: 4000 milliliters

Use the conversion relationship: 4 liters = 4000 milliliters.

6.20 Answer: 16000 pounds

Use the conversion relationship: 8 tons = 16000 pounds.

6.21 Answer: 175 cubic units

Multiply cubes in one layer by layers: $25 \times 7 = 175$ cubic units.

6.22 Answer: 208 cubic units

Multiply cubes in one layer by layers: $26 \times 8 = 208$ cubic units.

6.23 Answer: 243 cubic units

Multiply cubes in one layer by layers: $27 \times 9 = 243$ cubic units.

6.24 Answer: 280 cubic units

Multiply cubes in one layer by layers: $28 \times 10 = 280$ cubic units.

6.25 Answer: 319 cubic units

Multiply cubes in one layer by layers: $29 \times 11 = 319$ cubic units.

6.26 Answer: 360 cubic units

Multiply cubes in one layer by layers: $30 \times 12 = 360$ cubic units.

6.27 Answer: 403 cubic units

Multiply cubes in one layer by layers: $31 \times 13 = 403$ cubic units.

6.28 Answer: 448 cubic units

Multiply cubes in one layer by layers: $32 \times 14 = 448$ cubic units.

6.29 Answer: 495 cubic units

Multiply cubes in one layer by layers: $33 \times 15 = 495$ cubic units.

6.30 Answer: 544 cubic units

Multiply cubes in one layer by layers: $34 \times 16 = 544$ cubic units.

Chapter 7: Integers and the Coordinate Plane

Concept: Integers and absolute value

Integers include negative numbers, positive numbers, and zero. Absolute value is a number's distance from zero.

Questions

1. Find the absolute value of -15.
2. Find the absolute value of 8.
3. Find the absolute value of -3.
4. Find the absolute value of 0.
5. Find the absolute value of 22.
6. Find the absolute value of -11.
7. Find the absolute value of 14.
8. Find the absolute value of -9.
9. Find the absolute value of 6.
10. Find the absolute value of -20.
11. Find the absolute value of 17.
12. Find the absolute value of -4.
13. Find the absolute value of 25.
14. Find the absolute value of -7.
15. Find the absolute value of 12.

Concept: The coordinate plane

An ordered pair is written (x, y) . The signs of x and y determine which quadrant contains the point.

Questions

16. In which quadrant is the point $(-2, 4)$?
17. In which quadrant is the point $(3, 5)$?
18. In which quadrant is the point $(-4, -1)$?
19. In which quadrant is the point $(6, -3)$?
20. In which quadrant is the point $(-5, 2)$?
21. In which quadrant is the point $(1, -6)$?
22. In which quadrant is the point $(-3, -4)$?
23. In which quadrant is the point $(5, 1)$?
24. In which quadrant is the point $(-1, 7)$?
25. In which quadrant is the point $(4, -5)$?
26. In which quadrant is the point $(-6, 3)$?
27. In which quadrant is the point $(2, -2)$?
28. In which quadrant is the point $(-7, -3)$?
29. In which quadrant is the point $(8, 4)$?
30. In which quadrant is the point $(-2, -8)$?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

7.1 Answer: 15

Absolute value is distance from 0, so $|-15| = 15$.

7.2 Answer: 8

Absolute value is distance from 0, so $|8| = 8$.

7.3 Answer: 3

Absolute value is distance from 0, so $|-3| = 3$.

7.4 Answer: 0

Absolute value is distance from 0, so $|0| = 0$.

7.5 Answer: 22

Absolute value is distance from 0, so $|22| = 22$.

7.6 Answer: 11

Absolute value is distance from 0, so $|-11| = 11$.

7.7 Answer: 14

Absolute value is distance from 0, so $|14| = 14$.

7.8 Answer: 9

Absolute value is distance from 0, so $|-9| = 9$.

7.9 Answer: 6

Absolute value is distance from 0, so $|6| = 6$.

7.10 Answer: 20

Absolute value is distance from 0, so $|-20| = 20$.

7.11 Answer: 17

Absolute value is distance from 0, so $|17| = 17$.

7.12 Answer: 4

Absolute value is distance from 0, so $|-4| = 4$.

7.13 Answer: 25

Absolute value is distance from 0, so $|25| = 25$.

7.14 Answer: 7

Absolute value is distance from 0, so $|-7| = 7$.

7.15 Answer: 12

Absolute value is distance from 0, so $|12| = 12$.

7.16 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.17 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.18 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.19 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.20 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.21 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.22 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.23 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.24 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.25 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.26 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.27 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.28 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.29 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.30 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

Chapter 8: Advanced Multiplication and Division with Fractions and Decimals

Concept: Multiplying fractions by whole numbers

Multiply a fraction by a whole number, then simplify or write the answer as a mixed number when needed.

Questions

1. Find $\frac{1}{3} \times 6$.
2. Find $\frac{2}{5} \times 10$.
3. Find $\frac{3}{4} \times 8$.
4. Find $\frac{5}{6} \times 12$.
5. Find $\frac{3}{8} \times 16$.
6. Find $\frac{7}{10} \times 20$.
7. Find $\frac{4}{9} \times 18$.
8. Find $\frac{5}{12} \times 24$.

Concept: Dividing fractions by whole numbers

Dividing a fraction by a whole number means splitting it into that many equal groups; multiply by the reciprocal of the whole number.

Questions

9. Find $\frac{3}{4}$ divided by 2.
10. Find $\frac{5}{6}$ divided by 3.
11. Find $\frac{7}{8}$ divided by 4.
12. Find $\frac{2}{3}$ divided by 5.
13. Find $\frac{9}{10}$ divided by 3.

14. Find $5/12$ divided by 2.

15. Find $7/9$ divided by 7.

Concept: Long division and remainders

Divide one place at a time. Check a quotient and remainder with $\text{divisor} \times \text{quotient} + \text{remainder} = \text{dividend}$.

Questions

16. Divide 864 by 7. Give a quotient and remainder.

17. Divide 975 by 8. Give a quotient and remainder.

18. Divide 1,234 by 9. Give a quotient and remainder.

19. Divide 1,465 by 6. Give a quotient and remainder.

20. Divide 2,387 by 5. Give a quotient and remainder.

21. Divide 3,210 by 4. Give a quotient and remainder.

22. Divide 1,756 by 3. Give a quotient and remainder.

23. Divide 2,689 by 7. Give a quotient and remainder.

Concept: Decimal multiplication and division

Use place value carefully. Estimate first, then place the decimal point so the result is reasonable.

Questions

24. Find 2.40×3.50 .

25. Find 1.25×4.00 .

26. Find 3.60 divided by 0.50.

27. Find 7.20 divided by 0.30.

28. Find 4.08×2.00 .

29. Find 6.25 divided by 0.50.

30. Find 8.40×1.20 .

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

8.1 Answer: 2

Multiply the whole number: $\frac{1}{3} \times 6 = 2$.

8.2 Answer: 4

Multiply the whole number: $\frac{2}{5} \times 10 = 4$.

8.3 Answer: 6

Multiply the whole number: $\frac{3}{4} \times 8 = 6$.

8.4 Answer: 10

Multiply the whole number: $\frac{5}{6} \times 12 = 10$.

8.5 Answer: 6

Multiply the whole number: $\frac{3}{8} \times 16 = 6$.

8.6 Answer: 14

Multiply the whole number: $\frac{7}{10} \times 20 = 14$.

8.7 Answer: 8

Multiply the whole number: $\frac{4}{9} \times 18 = 8$.

8.8 Answer: 10

Multiply the whole number: $\frac{5}{12} \times 24 = 10$.

8.9 Answer: $\frac{3}{8}$

Dividing by 2 means multiplying by $\frac{1}{2}$: $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$.

8.10 Answer: $\frac{5}{18}$

Dividing by 3 means multiplying by $\frac{1}{3}$: $\frac{5}{6} \times \frac{1}{3} = \frac{5}{18}$.

8.11 Answer: $\frac{7}{32}$

Dividing by 4 means multiplying by $\frac{1}{4}$: $\frac{7}{8} \times \frac{1}{4} = \frac{7}{32}$.

8.12 Answer: $\frac{2}{15}$

Dividing by 5 means multiplying by $\frac{1}{5}$: $\frac{2}{3} \times \frac{1}{5} = \frac{2}{15}$.

8.13 Answer: $\frac{3}{10}$

Dividing by 3 means multiplying by $\frac{1}{3}$: $\frac{9}{10} \times \frac{1}{3} = \frac{3}{10}$.

8.14 Answer: $\frac{5}{24}$

Dividing by 2 means multiplying by $\frac{1}{2}$: $\frac{5}{12} \times \frac{1}{2} = \frac{5}{24}$.

8.15 Answer: $\frac{1}{9}$

Dividing by 7 means multiplying by $\frac{1}{7}$: $\frac{7}{9} \times \frac{1}{7} = \frac{1}{9}$.

8.16 Answer: 123 R 3

864 divided by 7 = 123 remainder 3, because $7 \times 123 + 3 = 864$.

8.17 Answer: 121 R 7

975 divided by 8 = 121 remainder 7, because $8 \times 121 + 7 = 975$.

8.18 Answer: 137 R 1

1,234 divided by 9 = 137 remainder 1, because $9 \times 137 + 1 = 1,234$.

8.19 Answer: 244 R 1

1,465 divided by 6 = 244 remainder 1, because $6 \times 244 + 1 = 1,465$.

8.20 Answer: 477 R 2

2,387 divided by 5 = 477 remainder 2, because $5 \times 477 + 2 = 2,387$.

8.21 Answer: 802 R 2

3,210 divided by 4 = 802 remainder 2, because $4 \times 802 + 2 = 3,210$.

8.22 Answer: 585 R 1

1,756 divided by 3 = 585 remainder 1, because $3 \times 585 + 1 = 1,756$.

8.23 Answer: 384 R 1

2,689 divided by 7 = 384 remainder 1, because $7 \times 384 + 1 = 2,689$.

8.24 Answer: 8.40

Compute the decimal operation: $2.40 \times 3.50 = 8.40$.

8.25 Answer: 5.00

Compute the decimal operation: $1.25 \times 4.00 = 5.00$.

8.26 Answer: 7.20

Compute the decimal operation: $3.60 \text{ divided by } 0.50 = 7.20$.

8.27 Answer: 24.00

Compute the decimal operation: $7.20 \text{ divided by } 0.30 = 24.00$.

8.28 Answer: 8.16

Compute the decimal operation: $4.08 \times 2.00 = 8.16$.

8.29 Answer: 12.50

Compute the decimal operation: $6.25 \text{ divided by } 0.50 = 12.50$.

8.30 Answer: 10.08

Compute the decimal operation: $8.40 \times 1.20 = 10.08$.